

Lab 2: Ratios, Proportionalities, and Units

The study of astronomy involves aspects of physics, chemistry, geology, and—you guessed it—lots and lots of math. But complicated equations don't have to stand in the way of your understanding of the Universe! Ratios and proportionalities can simplify the relationships expressed in complex equations. Because these equations relate measurable quantities, they have units associated with them. Keeping track of the units is key to understanding the Universe. In this lab you will use ratios, proportionalities, and unit analysis to simplify equations and solve problems.

Learning Objectives

At the completion of this lab, you should be able to:

- distinguish between constants and variables.
- cancel units in proportionalities.
- manipulate ratios and proportionalities.
- convert between different units of measure.

Constants and Variables

Equations demonstrate relationships between measured quantities using combinations of constants and variables. As you might be able to tell, constants have a fixed value while variables can change in value. Sometimes constants are numbers, but sometimes special numbers are represented by Roman or Greek letters, like c for the speed of light (300,000 kilometers per second) and π for 3.14159(...). Note that in equations, numbers or letters next to one another are multiplied together the multiplication symbol is implied.

In the equations below, circle the variables and cross out the constants.

$$C = 2\pi R$$

$$A = \pi R^2$$

$$E = mc^2$$

$$V = D/t$$

Unit Analysis

Measured quantities (like length, mass, time, money, etc.) have units associated with them (centimeters, grams, seconds, dollars, etc.). It is often necessary to change the units of a quantity in order to make it easier to deal with. Any known equality (e.g., $10^2 \text{ cm} = 1 \text{ m}$ or $1 \text{ inch} = 2.54 \text{ cm}$) can be made into a fraction that is equal to one, called a *conversion factor*. Because the conversion factor is equal to one, it can be multiplied by the quantity whose units need to be changed.

For example, how many quarters do you need to pay for a \$2.75 load of laundry?

What is the known quantity?

What is the conversion factor?

When a unit appears both in the top and bottom of the fraction, it can be canceled out, leaving you with only the desired units. You do this in your head every day (e.g. four quarters equals one dollar, or 60 minutes equals an hour), but it can be daunting when the units are unfamiliar or multiple conversion factors are required. If the conversion is approached carefully and carried out methodically, it's pretty simple.

A snail moves 1.5 meters every day. How many kilometers does the snail move in a year?

Start with the known quantity and units, then use conversion factors to cancel units and replace them with the units you want:

$$\frac{1.5 \cancel{m}}{1 \cancel{day}} \times \frac{1 \text{ km}}{10^3 \cancel{m}} \times \frac{3.65 \times 10^2 \cancel{days}}{1 \text{ year}} = \frac{5.5 \times 10^{-1} \text{ km}}{1 \text{ year}}$$

How many seconds are in one year? Keep track of all units! You will use this result in the next few questions.

A light-year is a *distance*, often used in astronomy news articles. Light travels 3×10^5 km in one second. How far does light travel in one year, in kilometers? This distance is a *light-year*.

The Sun's nearest neighbor, Alpha Centauri, is 4×10^{13} km away. How many years would it take to drive to Alpha Centauri at a freeway speed of 30 m/s (about 65 mph)? Express your answer in years using both scientific notation and written English (i.e., words!).

Ratios

When we want to compare two quantities with the same units, we can divide the two numbers and express the result as a *ratio*. For example, about 38 million people live in California, and about 19 million people live in Florida.

$$\frac{N_{\text{CA}}}{N_{\text{FL}}} = \frac{38,000,000}{19,000,000} = 2$$

The ratio of Californians to Floridians is about 38 million divided by about 19 million, or about 2; there are approximately twice as many people living in California as in Florida.

Ratios are very useful in astronomy because they can save a lot of work and aid in understanding. For example, Jupiter is 778 million kilometers from the Sun, or 7.78×10^8 km. It is hard to comprehend that distance because it is so much larger than anything familiar. Using a ratio makes the distance to Jupiter meaningful.

The Earth is 1.50×10^8 km from the Sun. How many times further away from the Sun is Jupiter than the Earth?

What are the units of a ratio?

Let's say we scaled down the Sun to the size of a basketball. Alpha Centauri would also be the size of a basketball, but how far away would it be at this scale? (Convert your answer to useful units!) The radius of the Sun is 7×10^9 cm, the radius of a basketball is about 12 cm, and the distance to Alpha Centauri is 4×10^{13} km.

Proportionalities

When the ratio involves an equation, you can cancel common variables, constants, and units to create a proportionality relation, greatly simplifying the problem.

For example, the volume of a sphere is $V = (4/3)\pi R^3$ so the volume is proportional to the radius cubed, or $V \propto R^3$, because $4/3$ and π are constants.

Sphere A has a radius twice as large as Sphere B. What is the ratio of volumes of the spheres?

Your weight is equal to your mass (a measure of the amount of *stuff* in something) times the acceleration of gravity, g . As a formula, that's:

$$W = mg$$

The acceleration of gravity on a planet depends on its mass, M , and its radius, R , like this:

$$g = \frac{GM}{R^2}$$

G is Newton's gravitational constant. Solve for the ratio of your weight on the Moon (W_M) compared to your weight on the Earth (W_E). **Do not use numbers yet.** Just use algebra to re-arrange the equations and cancel what you can.

The mass of the Earth is 82 times the mass of the Moon, and the radius of the Earth is 3.7 times the radius of the Moon. Find the numerical result for your weight on the Moon compared to your weight on the Earth.

How much lighter or heavier would you be on the Moon?